

The image only contains the information for $640 \times 480 = 307,200$ pixels, but we want to turn it into an image that contains $1024 \times 768 = 786,432$ pixels! Where are we going to get the information for all those extra pixels? The image editing program will now try to

“guess” these pixels based on the information it already has, and this is when things are apt to go horribly wrong. The image is going to pixelate, and the circle’s edges are going to end up jagged.



The Circle on the left is a vector; the one on the right is raster

In a vector graphic, as we discussed earlier, the final size of the circle is calculated on-the-fly, so no matter how much you resize a vector graphic, it’s going to have perfect, well-defined edges.

Vectors in Photoshop

In all fairness, Photoshop is hardly a program you would associate with vector graphics—for that, there are programs like Adobe Illustrator and CorelDRAW, both dedicated to the cause of vector art. However, Photoshop is quite capable in this department as well—it not only lets you easily create vector graphics, but also uses the principles of vectors to take its own raster editing capabilities one step ahead. There are three aspects of Photoshop that use vector graphics:

- Paths—the most direct way of creating and manipulating vectors in Photoshop. Paths are curves that are defined by key points, called *control points*, which can be manipulated at any time to change the shape of the curve.
- Type—starting with version 5.0, Photoshop treats text as a vector graphic, allowing you to resize and manipulate it for as long as you want before converting it into a raster.
- Shapes—to save the time it would take you to draw them using paths, Photoshop lets you draw basic shapes like rectangles, ellipses and polygons as vector images.